

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I B Tharun Kumar / 192424356(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B Tharun Kumar

Industry Problem Solving Task — Questions & Solutions

Question 1

A startup company with 20 employees is setting up a new office and needs a reliable, easy-to-manage LAN on a limited IT budget. As the network designer, propose a suitable network topology, specify the devices, cabling, and connections required, and justify your design choice.

Solution

1. Problem Analysis

The startup has a single-floor office with 20 users, standard requirements (internet access, file sharing, printing, VoIP calling) and no dedicated IT staff. The key constraints are: low upfront cost, simple day-to-day management, quick troubleshooting when a fault occurs, and headroom to add 5-10 more devices as the company grows. Because the business is not mission-critical (unlike a bank or hospital), a single point of failure at the core switch is an acceptable trade-off in exchange for simplicity and cost savings.

2. Recommended Topology & Justification

A Star topology is the most appropriate choice. Every device connects with its own dedicated cable to a central switch, so the network is easy

to monitor, wire, and expand. Most importantly, it isolates faults: if one workstation's cable or NIC fails, only that single node is affected while the remaining 19 employees continue working — unlike bus topology where one bad segment can bring down the whole LAN.

3. Network Devices Required

Device	Recommended Model / Type	Function
Core Switch	Cisco Catalyst 2960-24TT or Netgear GS724T (24-port, L2 managed, PoE+)	Central connection point for all workstations, printers, VoIP phones and the AP
Router / Firewall	Cisco RV340 or a pfSense appliance	Internet gateway, NAT, DHCP server, basic firewall/VPN

Device	Recommended Model / Type	Function
Wireless Access Point	Ubiquiti UniFi AP or TP-Link EAP	Wi-Fi coverage for laptops, phones and guest devices
IP Phones	Cisco/Yealink PoE VoIP handsets	Office telephony over the same LAN
End Devices	20 desktops/laptops + 2 network printers	User workstations and shared printing

4. Cabling & Transmission Media

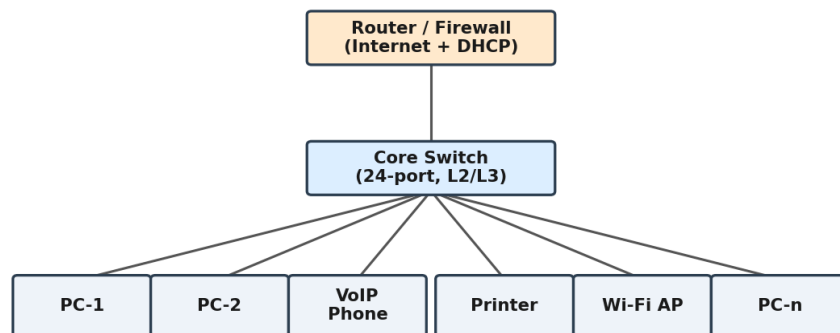
Every device is wired to the switch with Cat6 UTP cable terminated to the TIA/EIA-568-B standard using RJ45 connectors. Cat6 supports Gigabit Ethernet up to 100 metres, which comfortably covers a single office floor, and leaves headroom to run 10GBASE-T over shorter patch lengths if the company later upgrades its backbone. The switch

itself is PoE+, so the AP and IP phones draw power directly over the data cable, removing the need for separate power wiring at each mounting point.

5. IP Addressing Plan

A single subnet, 192.168.10.0/24, is sufficient for 20 users. The router's DHCP server leases addresses from 192.168.10.100–192.168.10.200, while the switch, AP, router and printers are given static reservations (192.168.10.1–192.168.10.20) so they are always reachable at the same address for management and print-queue configuration.

Fig 1. Star Topology — 20-Person Startup Office



All endpoints wired with Cat6 UTP (max 100 m) directly to the central switch
(single-point wiring = easy fault isolation & management)

Fig 1. Star topology — all endpoints home-run to a single core switch.

6. Tools & Technologies Used

- Cisco Packet Tracer — used to simulate the topology, verify IP addressing and DHCP scope before physical rollout
- Fluke LinkRunner / cable certifier — verifies every Cat6 run meets 1000BASE-T wiremap, length and crosstalk specifications
- PRTG / free SNMP monitor — polls the switch for port status and bandwidth utilisation

7. Testing & Validation

- Ping and traceroute between every host and the gateway to confirm end-to-end connectivity
- Cable certification report for each Cat6 run (length < 100 m, no wiremap faults)
- Fault-isolation test: unplug one workstation cable and confirm the remaining 19 devices are unaffected
- Throughput test (iperf3) between two hosts to confirm Gigabit line-rate is achieved
- VoIP call-quality test to ensure jitter/latency stay within acceptable limits when data and voice share the LAN

8. Why This Design Is Innovative & Scalable

Choosing a Layer 2/3 managed switch (rather than a basic unmanaged one) costs only slightly more but gives the startup the option to create VLANs later (e.g., separating guest Wi-Fi from the corporate LAN) without replacing hardware — future-proofing the investment while keeping the initial design simple and inexpensive.

Question 2

A bank's core operations demand near-zero downtime because even a few minutes of network outage can disrupt financial transactions and violate regulatory SLAs. Design a network topology that guarantees high availability, specifying the devices, cabling, and redundancy mechanisms, and justify your choice.

Solution

1. Problem Analysis

Banking systems process real-time transactions and must comply with strict regulatory uptime requirements (often 99.99%+). A single link or device failure must never take the whole network down. The design has to provide redundant paths at every layer, automatic failover in milliseconds, and low-latency, high-bandwidth links between the data centre and branch/ATM networks.

2. Recommended Topology & Justification

A full/partial Mesh topology at the core is the right choice. In a full mesh every core device — routers, core switches and firewalls — connects directly to every other core device, so no single link or node failure can isolate the network. For cost reasons, a bank typically implements a partial mesh: the core switches and firewalls are fully meshed with each other, while the distribution and access layers connect with redundant (dual) uplinks to at least two core devices. When a link fails, routing protocols reroute traffic through an alternate path within milliseconds, satisfying the bank's uptime and data-integrity requirements.

3. Network Devices Required

Device	Recommended Model / Type	Function
Core Switches (x2)	Cisco Catalyst 9500 series	Meshed core, inter-VLAN routing, high-speed backbone
Edge Routers (x2)	Cisco ASR or Juniper MX series	Dual, BGP-based redundant

Device	Recommended Model / Type	Function
		internet/WAN connectivity
Firewalls (x2, HA pair)	Palo Alto / Fortinet HA cluster	Perimeter security with automatic failover
Distribution Switches	Cisco Catalyst 9300 (dual-homed)	Connect branch/floor access layer to both core switches
Servers & Storage	Redundant servers with RAID arrays, dual PSUs, UPS	Core banking application and database hosting

4. Cabling & Transmission Media

All inter-core links use fiber optic cable: single-mode fiber for long-distance links between the data centre and branch buildings, and multi-mode OM4 fiber within the data centre itself. Fiber is immune to electromagnetic interference, supports very high bandwidth (10/40/100 Gbps), and keeps latency to a minimum — essential for

real-time financial transactions. Access-layer connections to end-user workstations still use Cat6/Cat6a UTP terminated per TIA/EIA-568-B.

5. Redundancy & Routing Mechanisms

- Dynamic routing protocols — OSPF within the campus and BGP for redundant internet/WAN paths — automatically reroute traffic around a failed link
- HSRP/VRRP on core switches so the default gateway fails over transparently if one core device goes down
- Dual power feeds, redundant PSUs and UPS/generator backup to protect against power loss
- RAID storage arrays and database replication/clustering to protect transaction data

Fig 2. Partial Mesh Topology — Bank Core Network

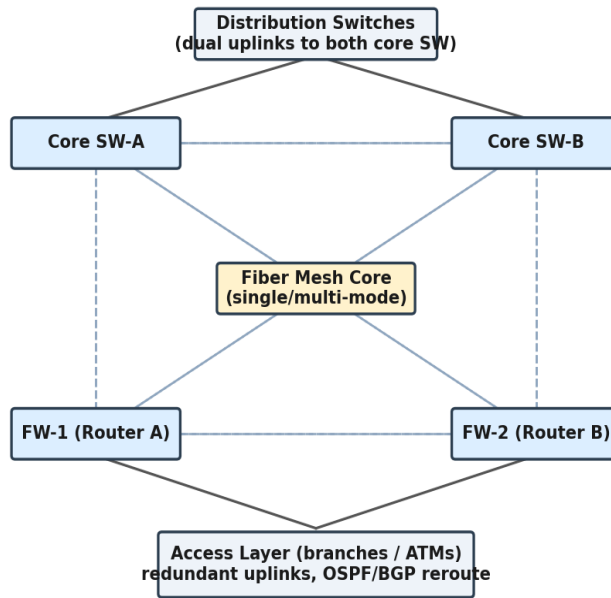


Fig 2. Partial mesh core with redundant, dual-homed distribution and access layers.

6. Tools & Technologies Used

- Cisco Packet Tracer / GNS3 — to model OSPF/BGP failover behaviour before deployment
- Wireshark — to analyse convergence time during a simulated link failure
- Network monitoring platform (SolarWinds / PRTG) with real-time alerting for link and device health

7. Testing & Validation

- Planned link-failure test: physically disconnect one core link and measure re-convergence time (target: sub-second)
- Failover test of the firewall HA pair and HSRP gateway to confirm zero packet loss during switchover
- Load/stress testing of transaction throughput during peak hours to confirm no bottlenecks
- Disaster-recovery drill: simulate a full data-centre outage and verify traffic shifts to the secondary site within the SLA window

8. Justification Summary

This mesh-based, fully redundant design directly satisfies the bank's near-zero-downtime requirement: there is no single point of failure at the core, traffic reroutes automatically in milliseconds, and every critical component (power, routing, storage, security) has a backup, which is exactly what banking-sector uptime and regulatory compliance demand.

Question 3

A school wants to set up a small computer lab of 10-15 machines for basic teaching (internet browsing, file sharing) but has a very tight budget. Design the lowest-cost network topology that meets these needs, specifying devices and cabling, and justify your choice.

Solution

1. Problem Analysis

The lab is small, usage is light (browsing, typing, occasional file transfer), and there is no requirement for high availability — a short outage during a lesson is an inconvenience, not a business risk. The overriding constraint is minimising hardware cost, since the school cannot justify a managed switch or structured cabling budget.

2. Recommended Topology & Justification

A Bus topology offers the lowest hardware cost because all computers share a single common communication line (the backbone), eliminating the need for a central switch or hub. This trade-off — susceptibility to collisions and the fact that a single cable break disrupts all connected machines — is entirely acceptable for a supervised classroom with light, non-critical usage.

3. Network Devices Required

Component	Specification	Purpose
Backbone cable	RG-58 coaxial, 50-ohm (classic) — or a single inexpensive 8/16-port	Shared communication line / bus backbone

Component	Specification	Purpose
	unmanaged switch (modern low-cost variant)	
T-connectors & BNC connectors	Standard 50-ohm BNC	Tap each PC onto the coaxial backbone
Terminators	50-ohm terminating resistors, one at each end of the bus	Absorb signal and prevent reflection/echo
Patch cables (modern variant)	Cat5e UTP, RJ45	Connect each PC to the low- cost unmanaged switch
Teacher's PC	Any standard desktop	Shares files/printer via built-in Windows/Linux sharing — no dedicated server needed

4. Cabling & Transmission Media

In the classic implementation, a single RG-58 coaxial cable runs the length of the room; each PC taps in via a T-connector, and 50-ohm terminators sit at both ends to prevent signal reflection. In the more contemporary low-cost implementation — recommended here for easier future migration — a single inexpensive 8- or 16-port unmanaged switch replaces the coaxial backbone/terminators, and Cat5e UTP patch cables (TIA/EIA-568-B, RJ45) connect each PC to the switch. This achieves bus-like simplicity and cost while remaining compatible with standard Ethernet NICs.

Fig 3. Bus Topology — Low-Cost Classroom Lab

Coaxial Backbone (RG-58) with T-connectors / BNC taps

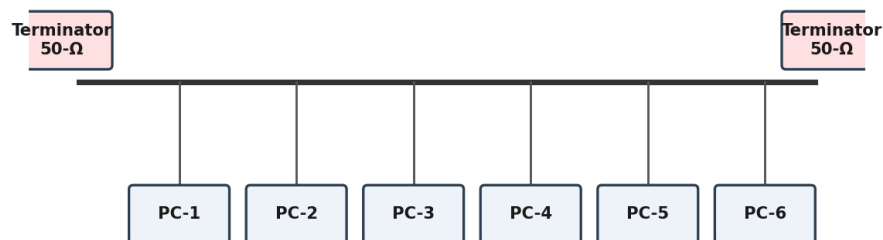


Fig 3. Bus topology — coaxial backbone with T-connectors and 50-ohm terminators.

5. Tools & Technologies Used

- BNC cable tester / continuity tester — verifies the coaxial run and terminator resistance

- Windows/Linux built-in file & printer sharing — avoids the cost of a dedicated file server
- Free bandwidth monitor (e.g., NetSpeedMonitor) to confirm the shared line isn't saturated during peak lab hours

6. Testing & Validation

- Continuity/terminator test at both ends of the coaxial run to confirm no open or short circuit
- Ping test from every PC to the teacher's PC and to the internet gateway
- Collision/load test with all machines browsing simultaneously to confirm acceptable performance for classroom use
- Fault-tolerance demonstration: disconnect one T-connector to show (and explain to students) that the whole segment goes down — reinforcing the topology's known limitation

7. Justification Summary

For a budget-constrained, low-criticality classroom lab, bus topology (or its modern unmanaged-switch equivalent) delivers functional internet and file-sharing access at the lowest possible hardware and cabling cost, which is precisely what the requirement calls for.

Question 4

A large university campus with multiple buildings (lecture halls, labs, library, dormitories, administration) needs a network serving thousands of users. Design a topology that provides redundancy at the backbone, simplicity within each building, and the ability to scale as new buildings are added. Justify your design.

Solution

1. Problem Analysis

A campus of this scale cannot be served by a single topology: the backbone needs redundancy against link failure, each building needs a simple, manageable wiring scheme, and individual labs need a low-cost solution. The design must also scale easily — adding a new building should not disrupt the existing network.

2. Recommended Topology & Justification

A Hybrid topology, built on the industry-standard three-layer hierarchical model (Core – Distribution – Access), combines the best properties of multiple designs: a fiber-based ring/partial-mesh at the campus backbone for redundancy, a star topology within each building for manageability, and simple bus/tree wiring within individual labs for cost efficiency.

3. Network Devices Required

Layer	Devices	Role
Core	High-speed fiber routers/multilayer switches in a ring or partial-mesh (e.g., Cisco Catalyst 9600)	Campus backbone; interconnects all buildings with redundant paths
Distribution	One distribution switch per building (dual fiber uplinks to the core)	Aggregates traffic from that building's floors/labs; connects to campus core
Access	Floor-level access switches (Cisco Catalyst 9200) + Wireless APs (PoE)	Connects individual devices, classrooms, dorm rooms and Wi-Fi

4. Cabling & Transmission Media

The campus core uses single-mode fiber optic cable running at 10 Gbps or 40 Gbps between buildings, arranged in a ring or partial mesh so that no single link failure isolates a building. Each building's distribution switch connects to the core via dual fiber uplinks for

redundancy. Inside each building, floor-level access switches connect to end devices using Cat6 UTP cable (TIA/EIA-568-B, max 100 m per run), and wireless access points are powered over Ethernet (PoE), eliminating separate power cabling.

Fig 4. Hybrid Topology — University Campus (Core / Distribution / Access)

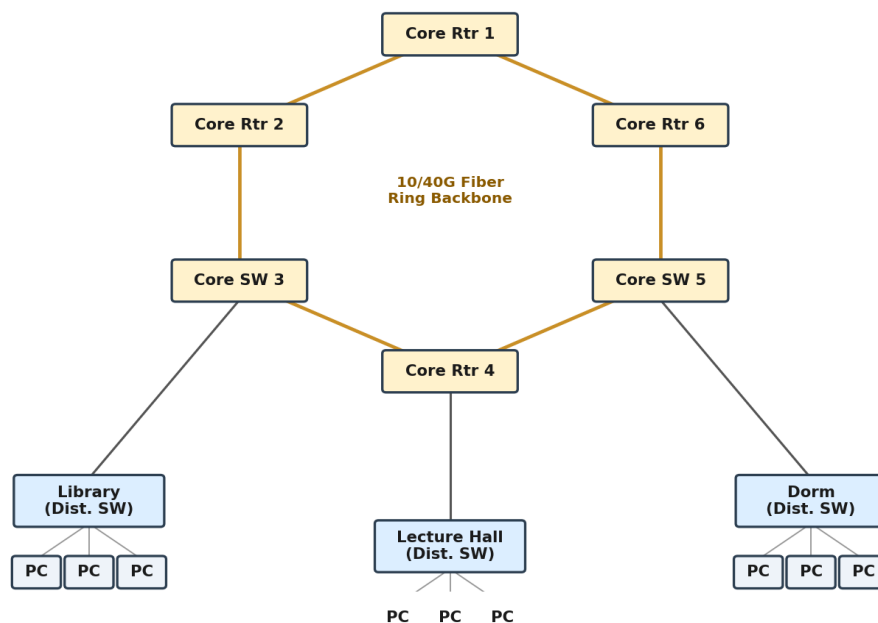


Fig 4. Hybrid topology — fiber-ring core, per-building star distribution, PoE wireless access.

5. Scalability

Because the hierarchical model cleanly separates core, distribution and access functions, adding a new building simply means installing a new distribution switch and running a fiber uplink to the core ring — with no disruption to existing buildings' traffic or wiring.

6. Tools & Technologies Used

- Cisco Packet Tracer / GNS3 — to model the ring/mesh backbone and simulate a link-failure reroute
- Fiber OTDR (Optical Time-Domain Reflectometer) — verifies fiber run quality and locates faults
- Centralised network management system (e.g., Cisco DNA Center) for campus-wide monitoring and PoE budget management

7. Testing & Validation

- Backbone failover test: disconnect one core fiber link and confirm traffic reroutes via the ring/mesh with no building isolated
- Per-building throughput test between the distribution switch and access layer during peak class hours
- Wireless coverage survey (site survey / heat map) across lecture halls, library and dormitories
- New-building onboarding drill: add a simulated distribution switch and confirm zero downtime for existing buildings

8. Justification Summary

The hybrid, three-layer design delivers redundancy at the backbone (ring/mesh core), simplicity at the building level (star distribution), and scalability campus-wide — exactly matching the needs of a large, multi-building university network.

Question 5

A mid-sized company has multiple departments — HR, Finance, Sales and IT — each requiring efficient internal communication but also strict logical separation from one another for security. Design a network architecture using switches and routers that meets these requirements, and justify your design.

Solution

1. Problem Analysis

Each department needs full connectivity for its own devices (PCs, printers) but must be logically isolated from other departments to improve security and reduce broadcast traffic — for example, Finance data should not be visible on the Sales network. The solution must also allow controlled inter-department communication (e.g., all departments reaching a shared file server) and provide secure internet access.

2. Recommended Architecture & Justification

A structured, scalable hierarchical architecture using VLAN-segmented access switches and a Layer 3 core switch is the right approach. Each department connects through its own dedicated Layer 2 access switch, forming the access layer; all access switches uplink to a central Layer 3 switch/router that performs inter-VLAN routing. VLANs logically separate each department's broadcast domain even though they may share the same physical switching fabric, giving both security and efficient use of cabling.

3. Network Devices & VLAN Plan

Department	Access Switch	VLAN ID	Subnet
HR	L2 access switch	VLAN 10	192.168.10.0/24
Finance	L2 access switch	VLAN 20	192.168.20.0/24
Sales	L2 access switch	VLAN 30	192.168.30.0/24
IT	L2 access switch	VLAN 40	192.168.40.0/24

All four access switches trunk (802.1Q) up to a central Layer 3 core switch, which performs inter-VLAN routing and applies access-

control lists (ACLs) to restrict which departments may talk to each other — for instance, allowing all VLANs to reach a shared print/file server while blocking direct Finance-to-Sales traffic. A perimeter router/firewall connects the core to the internet, with NAT and firewall rules protecting the internal network.

Fig 5. VLAN-Segmented Departmental Network

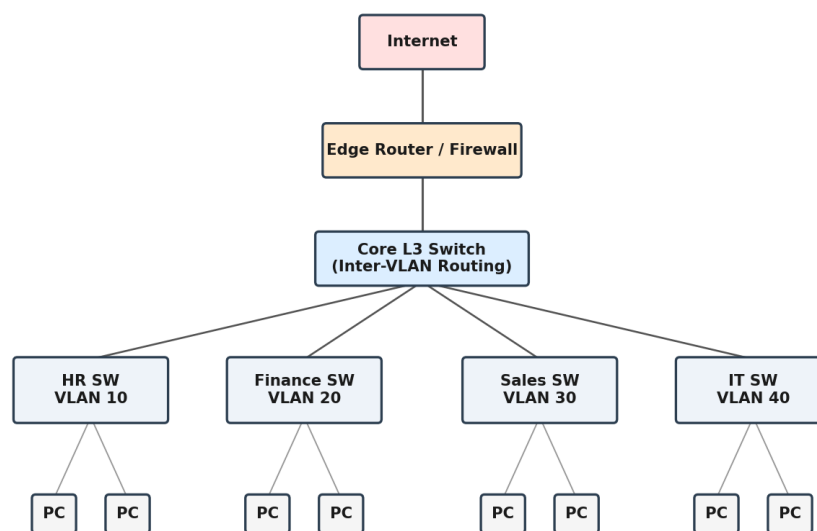


Fig 5. VLAN-segmented departmental network with Layer 3 inter-VLAN routing.

4. Cabling & Transmission Media

Access switches to end devices use Cat6 UTP (TIA/EIA-568-B, RJ45, up to 100 m). Uplinks from each access switch to the core switch use either Cat6a UTP or fiber (for longer runs between floors), configured as 802.1Q trunk links carrying all relevant VLAN tags.

5. Security & Reliability Enhancements

- ACLs / inter-VLAN routing rules on the core switch restrict traffic between departments to only what is explicitly required
- Redundant uplinks from each access switch to two core devices, with spanning-tree (RSTP) to prevent loops while providing a backup path
- A firewall with NAT and VPN capability protects the internal network from external threats
- Proper IP addressing and subnetting per department for efficient, easy-to-manage network administration

6. Tools & Technologies Used

- Cisco Packet Tracer — to design and simulate the VLAN and inter-VLAN routing configuration before deployment
- Wireshark — to verify VLAN tagging (802.1Q) and confirm broadcast traffic stays within its own VLAN
- Network access control / SNMP monitoring tool — to track switch port status and enforce port security per department

7. Testing & Validation

- Ping test within each VLAN to confirm intra-department connectivity

- Cross-VLAN ping test to confirm the ACLs correctly permit/deny inter-department traffic as designed
- Trunk-link verification (show vlan / show trunk) to confirm all VLANs are correctly tagged across uplinks
- Spanning-tree failover test: disconnect the primary uplink from an access switch and confirm the redundant path takes over

8. Justification Summary

This VLAN-based hierarchical design gives each department efficient internal communication, strong logical security separation, reduced broadcast traffic, and a clear path to add new departments or buildings by simply adding another access switch and VLAN — meeting all of the company's stated requirements.

RUBRICS FOR CALCULATION

Criteria	Weight age	Exemplar y (4)	Proficien t (3)	Developi ng (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understan ding with minor gaps	Basic understan ding; some key aspects missing	Poor or incorrect understan ding of the problem
Solution Design	20%	Innovative , practical, and industry- relevant solution	Effective solution with minor limitation s	Partially suitable solution	Inappropri ate or unclear solution

Criteria	Weight age	Exemplar y (4)	Proficien t (3)	Developi ng (2)	Beginning (1)
Technical Implement ation	20%	Fully functional, accurate, and well- executed implement ation	Mostly functional with minor errors	Partially implemen ted solution	Incomplet e or non- functional implement ation
Use of Tools & Technolog ies	10%	Appropria te and advanced use of relevant tools and technologi es	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem- Solving Approach	10%	Logical, systematic , and well-	Mostly logical approach	Somewha t disorgani	No clear approach

Criteria	Weight age	Exemplar y (4)	Proficien t (3)	Developi ng (2)	Beginning (1)
		structured approach		zed approach	
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originalit y	No creativity
Testing & Validation	5%	Thorough testing with real- world scenarios	Adequate testing	Minimal testing	No testing
Document ation & Presentati on	5%	Clear, profession al, and well- document ed	Good document ation	Basic document ation	Poor or no document ation

Criteria	Weight age	Exemplar y (4)	Proficien t (3)	Developi ng (2)	Beginning (1)
Teamwork / Collaborat ion	5%	Excellent coordinati on and contributi on (if team- based)	Good collaborat ion	Limited participati on	No collaborati on or contributi on